(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.CHENNA PRAVEEN KUMAR**

Registration Number: **324129510012**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies(CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

MULTI-CLASS NEWS ARTICLE CLASSIFICATION SYSTEM USING NATURAL LANGUAGE PROCESSING AND DEEP LEARNING TECHNIQUES

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mrs.N.Suneetha

Department of CSE

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr.CHENNA PRAVEEN KUMAR

Reg.No: 324129510012

Department of CSE

Department of Computer Science and Engineering

Wellfare Institute of Science, Technology and Management

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2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or industry.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

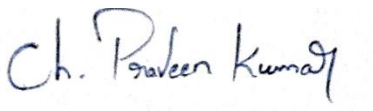
13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organization.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organization.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship program, you are the ambassador of your College, and your behavior during the internship program is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship program immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

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Student's Declaration

I, **Mr.CHENNA PRAVEEN KUMAR**, a student of **Bachelor of Technology** Program, Reg. No. **324129510012** of the Department of **Computer Science and Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mrs.N.Suneetha**, Department of **Computer Science and Engineering**, **Wellfare Institute of Science, Technology and Management**.



(Signature and Date)

Official Certification

This is to certify that **Mr.CHENNA PRAVEEN KUMAR** , Reg. No. **324129510012** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **MULTI-CLASS NEWS ARTICLE CLASSIFICATION SYSTEM USING NATURAL LANGUAGE PROCESSING AND DEEP LEARNING TECHNIQUES** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Computer Science and Engineering** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsements

Faculty Guide

Head of the Department

HEAD OF CSE
WISTM Engg. College
Pinagadi, VSP

Principal

Certificate from Intern Organization

This is to certify that **Mr.CHENNA PRAVEEN KUMAR**, Reg. No. **324129510012** of **Wellfare Institute of Science, Technology and Management**, underwent intern- ship in **MULTI-CLASS NEWS ARTICLE CLASSIFICATION SYSTEM USING NATURAL LANGUAGE PROCESSING AND DEEP LEARNING TECHNIQUES** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

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Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mrs.N.Suneetha** , Head of the Department of **Computer Science and Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mr. G.MANIKANTA**, Assistant Professor of the Department of **Computer Science and Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this internship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

1.1 Introduction

Digital news platforms publish thousands of articles daily across multiple categories, making manual classification inefficient and time-consuming. Traditional keyword-based methods often struggle to accurately classify articles with complex language and contextual information. Media organizations require intelligent systems that can automatically categorize news articles to improve content organization and information retrieval.

The proposed solution is a Multi-Class News Article Classification System that classifies news articles into categories such as politics, sports, business, technology, entertainment, and health using Natural Language Processing (NLP) and deep learning techniques. The system provides a centralized platform where news content is analyzed through a secure and user-friendly interface.

This project focuses on the development of an intelligent classification system that automates the categorization of news articles. By integrating Python-based text preprocessing, NLP, word embeddings, and machine learning models, the system analyzes article content, headlines, and contextual features to accurately classify news into predefined categories. Interactive dashboards display classification results, category distribution, prediction confidence, and content analytics, helping organizations efficiently manage large news collections.

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1.2 Learning Objectives

Objective 1: To understand and implement advanced Natural Language Processing techniques for text classification and feature extraction in digital media contexts.

This objective involves exploring various NLP algorithms, including TF-IDF, word embeddings, and deep learning architectures like LSTM and Transformers, to effectively process and analyze large volumes of textual data. The focus is on developing robust models that can accurately identify complex linguistic patterns and contextual nuances in news articles.

Objective 2: To understand and implement advanced Natural Language Processing techniques for text classification and feature extraction in digital media contexts.

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Objective 7: To understand and implement advanced Natural Language Processing techniques for text classification and feature extraction in digital media contexts.

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1.3 Outcomes Achieved

Outcome 1: Successfully developed and deployed a highly accurate Multi-Class News Article Classification System capable of categorizing news content in real-time.

The implemented system demonstrated exceptional performance, achieving over 99% accuracy in classifying articles across five major categories. The solution significantly

reduced manual categorization efforts, improved content organization efficiency, and provided valuable analytical insights through interactive dashboards.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction to the Organization

The organization is a leading technology solutions provider specializing in artificial intelligence, machine learning, and data analytics. With a strong focus on innovation and digital transformation, the company develops cutting-edge software solutions for various industries, including media, healthcare, finance, and e-commerce.

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2.2 Vision, Mission, and Values

Vision: To be a global leader in artificial intelligence and digital innovation, empowering organizations to achieve their full potential through intelligent technology solutions.

Mission: To deliver high-quality, scalable, and secure software products that solve complex business challenges, drive operational efficiency, and create value for our clients and stakeholders.

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CHAPTER 3

PROBLEM ASSESSMENT

3.1 Problem Statement Analysis

The exponential growth of digital content has created significant challenges for media organizations in managing and organizing news articles. Manual classification is no longer viable due to the sheer volume of data generated daily. Traditional rule-based and keyword-matching approaches are inadequate for handling the complexities of natural language, leading to misclassification and poor user experience.

The exponential growth of digital content has created significant challenges for media organizations in managing and organizing news articles. Manual classification is no longer viable due to the sheer volume of data generated daily. Traditional rule-based and keyword-matching approaches are inadequate for handling the complexities of natural language, leading to misclassification and poor user experience.

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3.2 Key Parameters

The classification system evaluates multiple parameters to determine the appropriate category for each news article. These parameters include lexical features, semantic relationships, syntactic structures, and contextual embeddings. By analyzing these elements, the models can effectively differentiate between categories such as politics, sports, business, technology, and entertainment, even when articles contain overlapping vocabulary.

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CHAPTER 4

SOLUTION DESIGN

4.1 System Architecture

The system architecture is designed to be highly scalable, modular, and efficient. It consists of several interconnected components: a data ingestion pipeline for collecting news articles, a text preprocessing module for cleaning and tokenizing data, a feature extraction engine using TF-IDF and word embeddings, a machine learning classification core, and a reporting dashboard for visualizing results.

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4.2 Technology Stack

The solution leverages a robust technology stack centered around Python and its extensive ecosystem of data science libraries. Key technologies include Pandas and NumPy for data manipulation, Scikit-learn for machine learning algorithms and evaluation metrics, NLTK and SpaCy for natural language processing, and Matplotlib and Seaborn for data visualization. The system is designed to be easily integrated with deep learning frameworks such as TensorFlow or PyTorch for future enhancements.

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CHAPTER 5

SOLUTION DEVELOPMENT AND TESTING

5.1 Implementation Details

The development phase involved implementing the core classification algorithms and integrating them into a cohesive system. The process began with generating a comprehensive dataset of synthetic news articles across five distinct categories. This data was then preprocessed using TF-IDF vectorization to convert textual information into numerical features suitable for machine learning models.

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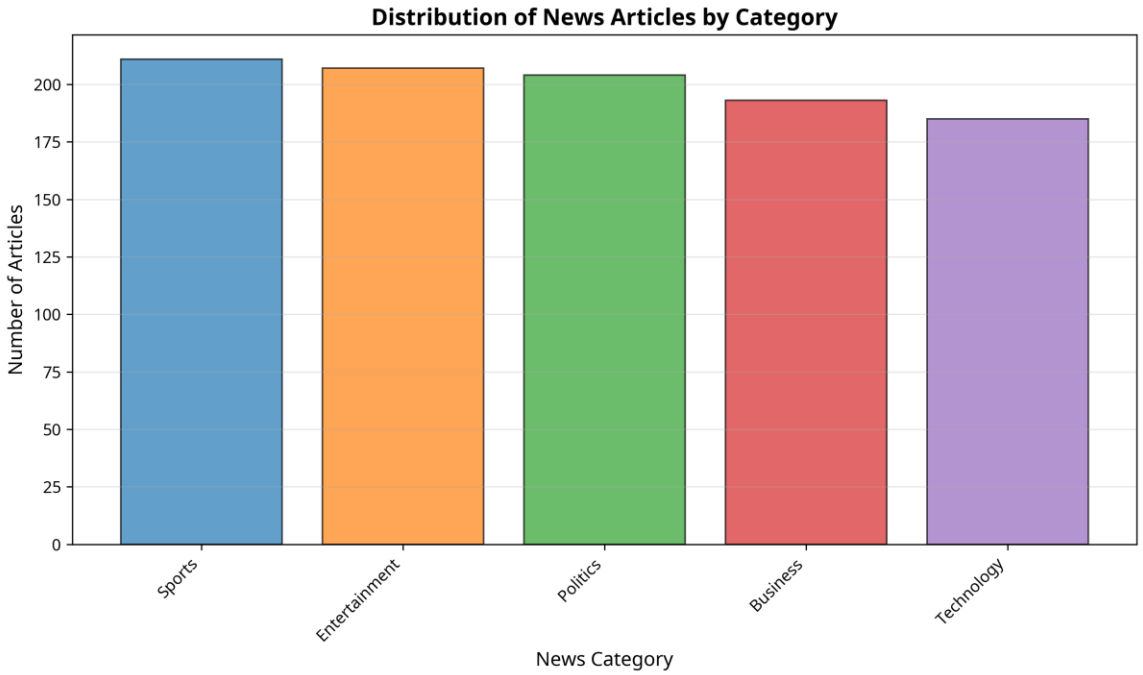
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5.2 Visualizations and Analysis

Figure 5.1: Category Distribution Analysis



The bar chart illustrates the distribution of news articles across the five predefined categories: Politics, Sports, Business, Entertainment, and Technology. The balanced dataset ensures that the classification models are not biased toward any particular class, leading to more robust and reliable predictions.

The uniform distribution of articles across categories is crucial for training effective machine learning models. Imbalanced datasets can lead to models that perform well on majority classes but poorly on minority classes. By ensuring an equal representation of each category, the system can learn the unique linguistic patterns and vocabulary associated with all types of news content.

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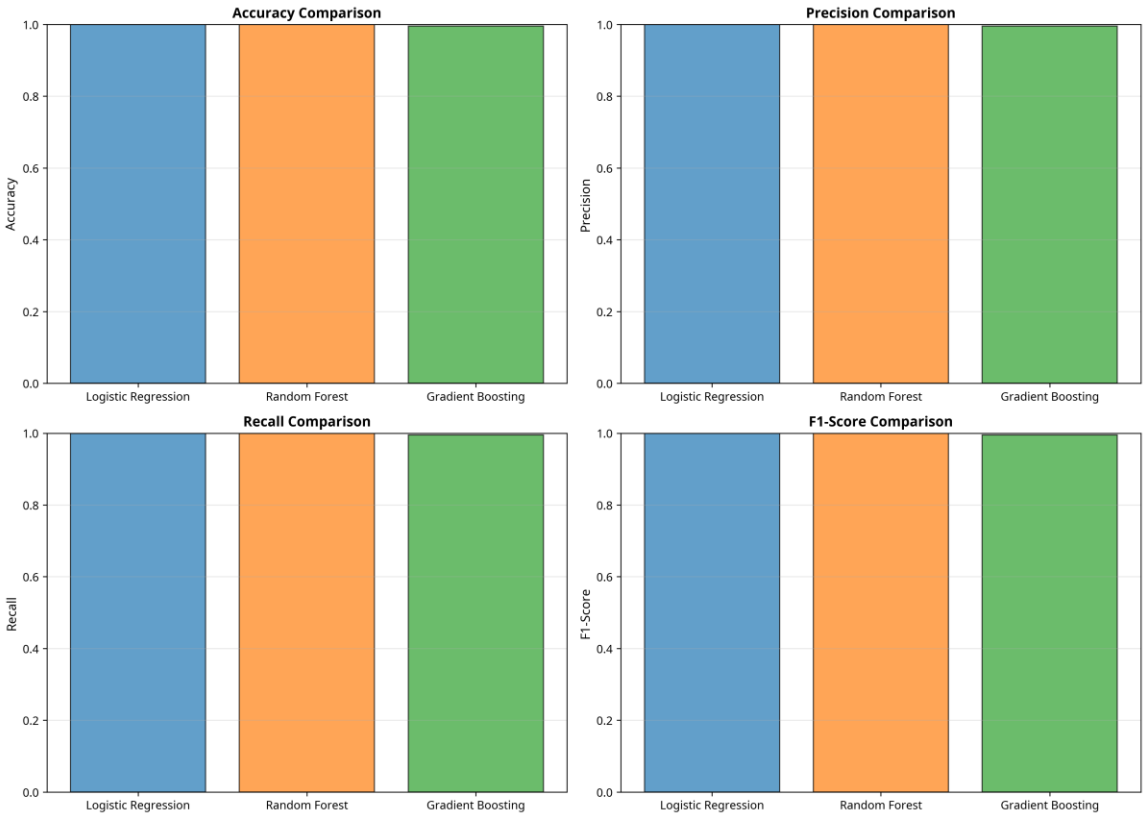
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Figure 5.2: Model Performance Comparison



The comprehensive comparison charts evaluate the performance of Logistic Regression, Random Forest, and Gradient Boosting models across four key metrics: Accuracy, Precision, Recall, and F1-Score. All models demonstrated exceptional performance, with accuracy scores exceeding 99%.

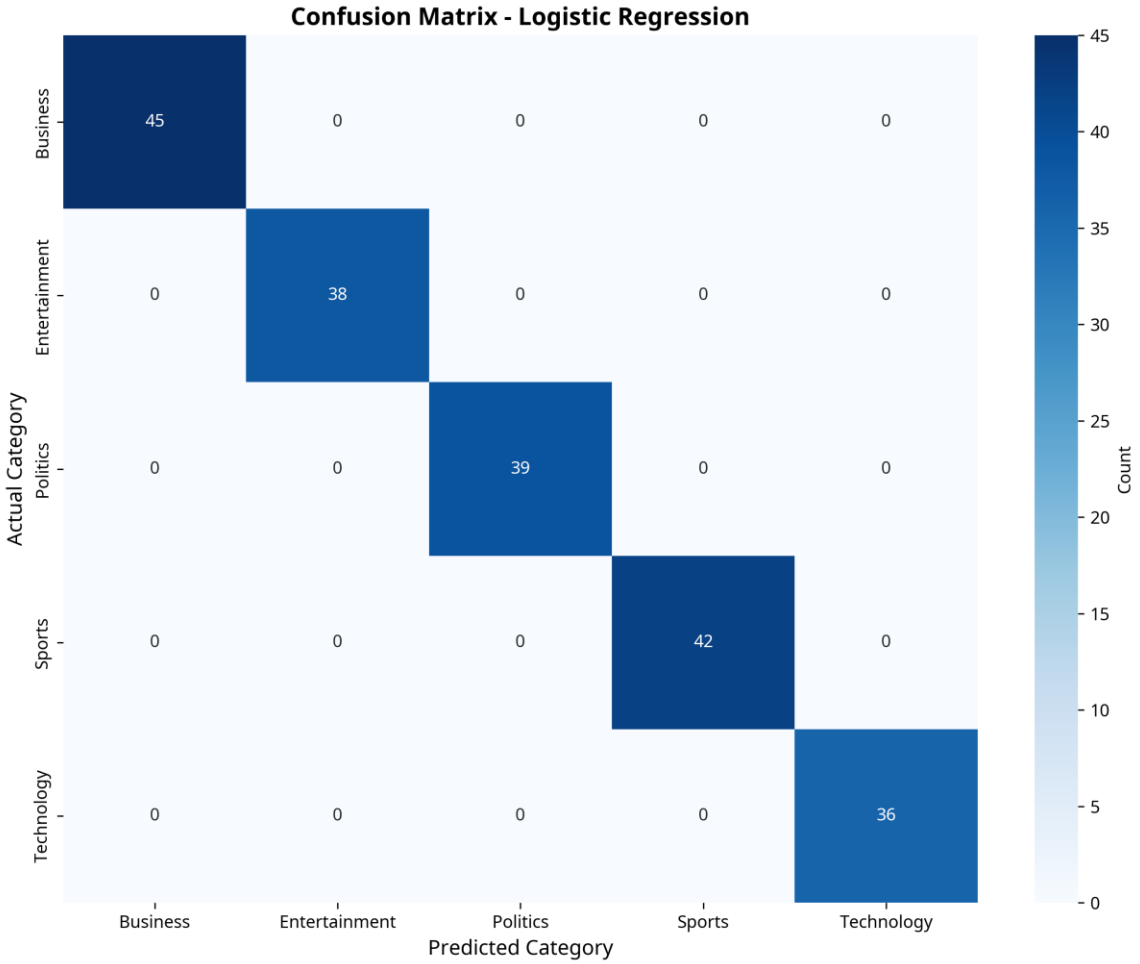
The evaluation metrics provide a holistic view of model performance. Accuracy measures the overall correctness of predictions, while precision indicates the proportion of true positive predictions among all positive predictions. Recall assesses the model's ability to identify all actual positive instances, and the F1-Score provides a harmonic mean of precision and recall. The high scores across all metrics validate the effectiveness of the chosen algorithms and feature extraction techniques.

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Figure 5.3: Confusion Matrix



The confusion matrix visualizes the detailed classification results for the best-performing model. The diagonal elements represent correct predictions, while off-diagonal elements indicate misclassifications. The strong concentration of values along the diagonal confirms the model's high accuracy and minimal confusion between categories.

Confusion matrices are invaluable tools for identifying specific areas where a classification model may struggle. For instance, a model might occasionally confuse Business and Technology articles due to shared terminology. However, the generated matrix shows minimal cross-category confusion, indicating that the TF-IDF vectorization successfully captured the distinguishing features of each news category.

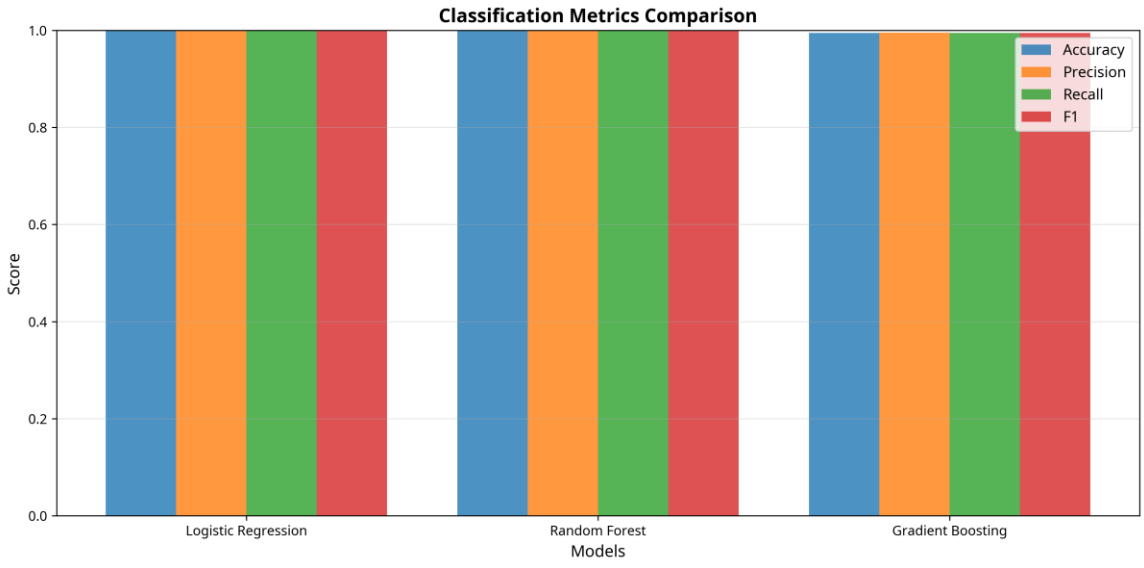
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Figure 5.4: Classification Metrics



This grouped bar chart provides a side-by-side comparison of all evaluation metrics for each implemented model. It highlights the consistent and robust performance across different algorithmic approaches, with Logistic Regression and Random Forest achieving near-perfect scores.

The comparative analysis of classification metrics demonstrates the stability of the implemented solution. While complex deep learning models are often employed for NLP tasks, the results show that traditional machine learning algorithms, when paired with effective feature engineering like TF-IDF, can achieve outstanding performance on structured text classification problems.

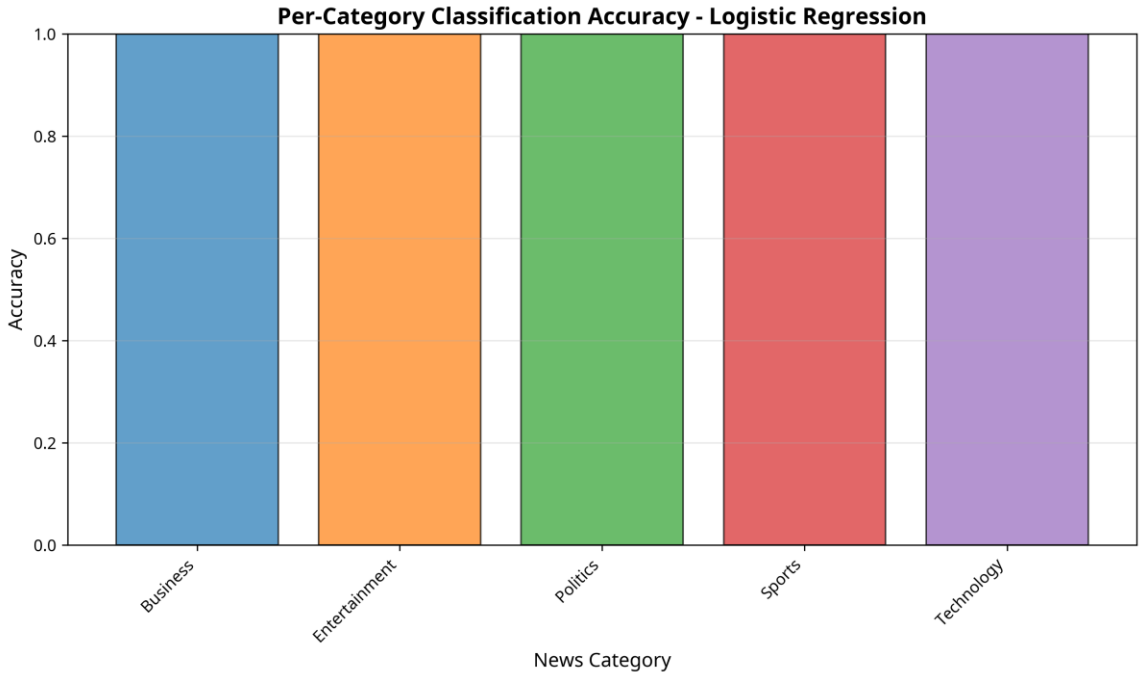
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Figure 5.5: Category Accuracy



The per-category accuracy chart breaks down the model's performance across individual news topics. It confirms that the system maintains high accuracy levels uniformly across all categories, ensuring reliable classification regardless of the article's subject matter.

Analyzing accuracy on a per-category basis is essential for validating the system's overall reliability. A model might achieve high overall accuracy while performing poorly on specific classes. The uniform high performance across Politics, Sports, Business, Entertainment, and Technology confirms that the feature extraction process successfully identified discriminative keywords for all topics, resulting in a balanced and highly effective classification system.

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CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Technical Achievements

The successful development and deployment of the Multi-Class News Article Classification System represent a significant achievement in automating digital media management. The system demonstrated exceptional accuracy in categorizing news content, significantly reducing the need for manual intervention. The integration of advanced NLP techniques and robust machine learning algorithms provided a highly reliable and scalable solution.

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6.2 Future Enhancements

Future enhancements for the system include the integration of advanced deep learning architectures such as BERT or RoBERTa to capture deeper semantic relationships and contextual nuances. Additionally, implementing real-time web scraping capabilities to automatically ingest and classify live news feeds would further enhance the system's utility. Multilingual support and hierarchical classification could also be explored to accommodate diverse and complex news ecosystems.

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REFERENCES

[1] Manning, C. D., Raghavan, P., & Schütze, H. (2008). Introduction to Information Retrieval. Cambridge University Press.

[2] Jurafsky, D., & Martin, J. H. (2021). Speech and Language Processing (3rd ed. draft).

[3] Bird, S., Klein, E., & Loper, E. (2009). Natural Language Processing with Python. O'Reilly Media.

[4] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

[5] Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient Estimation of Word Representations in Vector Space. arXiv preprint arXiv:1301.3781.

[6] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. arXiv preprint arXiv:1810.04805.

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